

Multiple Choice (10 marks)

1. In Python 3, `b = [11,13,15,17,19,21]; print(b[::2])` outputs what?
19,21
11,15
11,15,19
13,17,21
2. Which of the following sorting algorithms has average-case and worst-case running times of $O(n \log n)$?
(a) Bubble sort
(b) Insertion sort
(c) Merge sort
(d) Quicksort
3. Which of the following decimal numbers has an exact representation in binary notation?
(a) 0.1
(b) 0.2
(c) 0.3
(d) 0.5
4. Which of the following is the name of the data structure in a compiler that is responsible for managing information about variables and their attributes?
(a) Abstract Syntax Tree (AST)
(b) Attribute Grammar
(c) Symbol Table
(d) Semantic Stack
5. Which of the following is NOT a property of bitmap graphics?
(a) Fast hardware exists to move blocks of pixels efficiently.
(b) Realistic lighting and shading can be done.
(c) All line segments can be displayed as straight.
(d) Polygons can be filled with solid colors and textures.

True / False (10 marks)

Answer True or False

6. True or False: The method 'sort()' can be called directly on a string object in Python to sort its characters.
7. True or False: The values of local variables are typically included in a subroutine's activation record frame in stack-based programming languages.
8. True or False: The sum of the list $l = [1, 2, 3, 4]$ in Python 3 is 8.
9. True or False: The one-time pad is considered a perfectly secure encryption scheme when used correctly, due to its use of truly random keys.
10. True or False: In Python 3, the expression $r(3)$ evaluates to 3 when r is defined as a lambda function multiplying its input by 2.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to count the number of digits of a given number.
12. Write a python function to count the number of pairs whose sum is equal to 'sum'.

End of Paper